23001951

SCS 3316 / IS 3215 - Software Quality Assurance

Practical 01

Part A

A1

a. Error
b. Fault
c. Failure
d. Fault
e. Fault

A2

Prevention (SQA)	Detection (QC / Testing)
Coding standards	Executing test cases
Requirements review	Load testing the finished build
Staff training	
Process audit	

Part B

B1

Problem	Environment characteristic that fixes it
1. Testing is performed on the live system using real student accounts.	Isolation: Use a separate test environment instead of the live system.
2. Tests cannot be repeated because course seats have already been taken.	Repeatability: Use controlled test data and reset the environment before each test.
3. The team does not test what happens when 2,000 students register together.	Realism and scalability: The test environment should simulate realistic workloads and support load testing.
4. Nobody can identify why a test failed or which build was tested.	Observability and traceability: Keep logs and record the software build and configuration used.

B2

1. Test data
2. Environment
3. Environment
4. Requirement and schedule

B3

Type	Test values	Reason
Normal	5, 12, 20	Valid values within the allowed range.
Boundary	0, 1, 2, 23, 24, 25	Tests the minimum and maximum values and the values immediately outside the range.
Invalid	-1, 0, 25, -5	These values are outside the allowed range.

Part C

C1

	Factor	Category
1	Efficiency	Product Operation
2	Integrity	Product Operation
3	Flexibility	Product Revision
4	Testability	Product Revision
5	Interoperability	Product Transition
6	Reliability	Product Operation

C2

Vague requirement	Measurable requirement	How we check
The mobile app should be fast.	The app shall respond to user requests within 2 seconds for at least 95% of requests under normal conditions.	Measure response times using performance testing.
The system must be secure.	The system shall block 100% of unauthorized access attempts in the defined security test cases.	Perform security testing using unauthorized access attempts.
New staff should find it easy to use.	At least 90% of new staff shall complete the main task within 5 minutes without assistance.	Ask new staff to perform the task and measure completion time and success rate.